



Department of Computer Science
Second Semester Project Report 2025/2026

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Title of Project: Flight Delay Prediction

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Abstract

Flight delays remain a major ongoing challenge in the global aviation industry, impacting airlines, airports, and thousands of passengers annually. This project aims to build an integrated system for analyzing and predicting airport arrival delays. Using a refined dataset combining daily on-time arrival records from Eurocontrol and OurAirports metadata, this phase focuses on developing and comparing two supervised regression models—linear regression and a decision tree—capable of predicting the average arrival delay for each flight at a given airport on a given day. The final dataset comprises 997,537 daily airport records across 43 countries from 2019 to 2026, with engineered characteristics including month, day of the week, season, flight volume, and country. The linear regression model achieved a coefficient of determination (R^2) of 0.0419 on the test set, while the decision tree model achieved an R^2 of approximately 0.0755, indicating that the nonlinear model accounts for a larger proportion of the variance in mean delay using the same characteristics. The results highlight the importance of operational, temporal, and geographic characteristics in interpreting delay patterns, as well as the limitations of the dataset, particularly the lack of real-time weather data and detailed operational cause data. The project concludes with a comparison of the two models, an interpretation of the inferred linear regression coefficients, and recommendations for future improvements.

Introduction

Flight delays are a costly and disruptive problem in the aviation industry. These delays result from a range of factors, including weather conditions, airport congestion, air traffic management restrictions, and operational issues at airlines and airports. Even approximate delay forecasting is valuable for airlines planning their schedules, airports managing their capacity, and passengers planning their travels.

This project uses a data science methodology to study arrival delays at airports across Europe, North Africa, and parts of the Caucasus, drawing on the Eurocontrol airport arrival delay dataset. The third phase of the project focused on collecting,

cleaning, and analyzing this data. The fourth stage, which this report covers, focuses on building and comparing two supervised machine learning models linear regression and decision tree—both covered in the training course, to predict the average arrival delay per flight at the daily airport level, and on critically evaluating the quality of each model.

...If we have basic information about the record of a particular airport's day (month, day of the week, year, season, country, and number of arriving flights), how accurate are our predictions of the average delay per flight? And does the nonlinear model significantly outperform the linear model in this task? As the results below show, both models extract a real but limited signal from these characteristics, with the decision tree model capturing a much larger portion of the underlying pattern than linear regression.

Background:

Understanding flight schedules and delays is essential in the aviation industry. Departure and arrival times define when a flight is supposed to leave and reach its destination whenever a flight does not follow this schedule, a delay occurs, in this project delays are measured as the average delay in minutes per airport per day, calculated from Air Traffic Flow Management records.

Airport factors like (location, size, and traffic volume) play a major role in delay patterns since busy airports with high traffic are more likely to experience bottlenecks and capacity issues, operational constraints such as demand capacity imbalances and air traffic control restrictions are also recorded to understand why delays occur we use this airport and operational data as inputs for machine learning models to predict the average arrival delay, our main models are Linear Regression and Decision Tree Regressor chosen because our target is a continuous numeric value rather than a simple yes OR no outcome.

Domain Explanation:

The aviation sector combines airports, flights, and daily schedules. When these elements don't align due to high demand or operational constraints, flights are delayed. This project focuses entirely on airport operations and air traffic management delay records, sourced from Eurocontrol, which cover more than 40 European countries over many years.

Airports vary in location, size, and congestion level. Larger or busier airports typically experience more delays due to high air traffic volume and limited capacity. Our data records the number of arriving flights, the number of delayed flights, and the total air traffic management delay minutes for each airport daily. The main objective is to analyze these patterns and build a model that predicts the average arrival delay in minutes.

Algorithm Background:

In this project our goal is to predict the average arrival delay in minutes per airport per day, since the target variable is continuous and numeric this is a regression problem rather than a classification problem we trained and compared two models Linear Regression as a simple interpretable baseline and Decision Tree Regressor to capture non linear relationships in the data.

Linear Regression assumes a linear relationship between input features and the target variable and is fast to train and easy to interpret, Decision Tree Regressor learns decision rules by recursively splitting the data and can capture complex interactions between features we limited the tree depth to six levels to prevent overfitting, both models were evaluated using MAE, RMSE, and R^2 , and Decision Tree outperformed Linear Regression with a test R^2 of 0.0755, explaining approximately 7.5% of the variance in average delay minutes.

Literature Review: Review of Related Research (Last Five Years)

Paper 1

Title: Flight Delay Prediction: A Dissecting Review of Recent Studies Using Machine Learning

Authors: Sebastian Wandelt, Xinyue Chen, Xiaoqian Sun

Published in: IEEE Transactions on Intelligent Transportation Systems (2025)

Summary: This paper reviews a large number of studies that used machine learning to predict flight delays. The authors looked at how different research teams collected their data, selected features, built models, and evaluated results. They found that flight delays are a worldwide problem caused by weather, infrastructure issues, and increasing air traffic. The paper confirms that machine learning models are widely used in this area and can produce good results, but challenges like data quality and model consistency still exist. The authors suggest that future work should focus on better data integration and more reliable evaluation methods.

Key Insight: This review confirms that our project is working on a well-known and important problem, and that using structured operational data is a solid approach for building delay prediction models.

Citation: Wandelt, S., Chen, X., & Sun, X. (2025). Flight Delay Prediction: A Dissecting Review of Recent Studies Using Machine Learning. IEEE Transactions on Intelligent Transportation Systems.

[Flight Delay Prediction: A Dissecting Review of Recent Studies Using Machine Learning | IEEE Journals & Magazine | IEEE Xplore](#)

Paper 2

Title: Machine Learning Approach for Flight Departure Delay Prediction and Analysis

Authors: Ehsan Esmaeilzadeh, Seyedmirsajad Mokhtarimousavi

Published in: Transportation Research Record (2020)

Summary: This study looked at what causes flight departure delays and tried to predict them using a Support Vector Machine model. The data came from three major airports in New York over a 20-day period. The researchers found that factors like pushback delay, taxi-out time, and the gap between demand and airport capacity had the strongest impact

on delays. Because SVM does not directly explain its decisions, they used sensitivity analysis to understand which variables mattered most.

Key Insight: This paper shows that operational factors like capacity imbalance and ground delays are strong predictors of flight delays, which is directly relevant to our project since we also use operational and traffic data from EUROCONTROL.

Citation: Esmaeilzadeh, E., & Mokhtarimousavi, S. (2020). Machine Learning Approach for Flight Departure Delay Prediction and Analysis. Transportation Research Record.

[Machine Learning Approach for Flight Departure Delay Prediction and Analysis - Ehsan Esmaeilzadeh, Seyedmirsajad Mokhtarimousavi, 2020](#)

Paper 3

Title: Applying Machine Learning to Aviation Big Data for Flight Delay Prediction

Authors: Yushan Jiang, Yongxin Liu, Dahai Liu, Houbing Song

Published in: IEEE DASC/PiCom/CBDCoM/CyberSciTech Conference (2020)

Summary: This paper combined two large datasets, one with airline performance records and another with weather and climate data, to predict flight arrival delays. The researchers tested several machine learning models including neural networks and multilayer perceptron. Their best model reached around 89% accuracy, showing that combining different types of data leads to better predictions. The study also emphasized that weather information adds a lot of value when predicting delays.

Key Insight: This paper highlights the benefit of combining flight and environmental data. In our project, we did not have weather data available, but this study motivates future improvements by showing how much accuracy can improve when weather features are included.

Citation: Jiang, Y., Liu, Y., Liu, D., & Song, H. (2020). Applying Machine Learning to Aviation Big Data for Flight Delay Prediction. IEEE DASC/PiCom/CBDCoM/CyberSciTech Conference.

[Applying Machine Learning to Aviation Big Data for Flight Delay Prediction | IEEE Conference Publication | IEEE Xplore](#)

Paper 4

Title: Flight Delay Prediction Based on Machine Learning

Authors: Irmak Hatipoğlu, Ömür Tosun, Nedret Tosun

Published in: LogForum Journal (2022) DOI: 10.17270/J.LOG.2022.655

Summary: This study used advanced machine learning models including XGBoost, LightGBM, and CatBoost to predict whether a flight would be delayed. The dataset included over 18,000 flight records from one airline over a full year. One important challenge the researchers faced was class imbalance, since delayed flights were much fewer than on-time flights. They handled this using a technique called SMOTE to balance the dataset before training. The results showed high prediction accuracy and confirmed that proper preprocessing makes a big difference in model performance.

Key Insight: This paper shows that cleaning and preparing data carefully before training is just as important as choosing the right model. It supports the preprocessing steps we followed in our project to make sure the data was ready for modeling.

Citation: Hatipoğlu, I., Tosun, Ö., & Tosun, N. (2022). Flight Delay Prediction Based on Machine Learning. LogForum, 18(1). DOI: 10.17270/J.LOG.2022.655

[Flight delay prediction based with machine learning - LogForum - Tom Vol. 18, no. 1 \(2022\) - BazTech - Yadda](#)

Paper 5

Title: Flight Delay Prediction Using Machine Learning and Data Mining Approaches

Authors: Not specified in the available source

Published in: Academic Journal (covers U.S. and Iranian flight networks)

Summary: This paper tested several machine learning and data mining methods to predict flight delays, including Decision Trees, Random Forest, Bayesian models, and clustering approaches. The researchers used real flight data from both the United States and Iran, which made the study interesting because the two networks have very different characteristics. In the U.S. data, visibility, wind conditions, and departure time were the most important features. In the Iranian data, aircraft type and fleet age mattered more. All models reached above 70% accuracy, showing that machine learning works well across different aviation contexts.

Key Insight: This paper shows that the most important features for predicting delays can vary depending on the region and data source. This is a useful reminder that feature selection should always be guided by the specific dataset being used, just as we did in our project by focusing on EUROCONTROL operational features.

Citation: Author(s) not identified. Flight Delay Prediction Using Machine Learning and Data Mining Approaches. Available at: Airline-delay-prediction-by-machine-learning-algorithms.pdf

[Airline-delay-prediction-by-machine-learning-algorithms.pdf](#)

Dataset Description

The dataset used in this project is the cleaned version of the EUROCONTROL Airport Arrival ATFM Delay dataset produced in Phase 3. Each row represents a single airport on a single day, and includes the number of arriving flights, the total ATFM (Air Traffic Flow Management) arrival delay in minutes broken down by cause code, the number of delayed flights, and the number of flights delayed by more than 15 minutes.

Key characteristics of the cleaned dataset:

- 997,537 rows (airport-day records)
- 43 countries represented, including Belgium, Germany, France, the United Kingdom, Spain, Italy, Türkiye, Israel, Morocco, and others
- 405 unique airports identified by ICAO code
- Date range from January 2019 to April 2026
- 28 columns after cleaning, including delay-cause breakdowns (DLY_APT_ARR_A_1 through DLY_APT_ARR_W_1), flight counts, and engineered features (season, weekday)

For this phase, a derived target variable, `avg_delay`, was created by dividing the total arrival delay (`DLY_APT_ARR_1`) by the number of arriving flights (`FLT_ARR_1`) for each airport-day record. This produces a continuous numeric target representing the average ATFM delay per flight, in minutes, which is suitable for a regression task. The distribution of this target is heavily skewed: the mean is approximately 0.20 minutes, the 75th percentile is 0, and the maximum is 1,032 minutes, reflecting the fact that the large majority of airport-days experience no measurable ATFM delay while a small number experience severe disruption.

Data Cleaning

The raw EUROCONTROL export required several cleaning steps before it could be used for modeling. These steps, completed prior to this phase, included:

1. Removing fully empty columns (e.g., an unnamed trailing column with no data).
2. Dropping columns that provided no analytical value, such as the internal `ATFM_VERSION` tag and the redundant Pivot Label text field (airport name and ICAO code combined into one string).

3. Converting all delay-cause columns (DLY_APT_ARR_A_1 through DLY_APT_ARR_NA_1), which were inconsistently typed as text (object) due to embedded formatting artifacts, into numeric (float) columns using `pandas.to_numeric` with error coercion.
4. Replacing missing values in the delay columns with 0, reflecting the fact that a missing delay entry corresponds to no recorded delay for that cause on that day, rather than an unknown value.
5. Verifying that no negative delay values, impossible flight counts, or invalid years existed in the cleaned data.
6. Removing duplicate rows.

After cleaning, the dataset contained 997,537 rows and 28 columns with zero remaining missing values, confirmed using `df.isnull().sum()`.

Preprocessing for Modeling

Building on the cleaned dataset, the following preprocessing steps were applied specifically to prepare the data for the Linear Regression model:

Target Variable Construction

The target variable, `avg_delay`, was computed as the total arrival delay (DLY_APT_ARR_1) divided by the number of arriving flights (`FLT_ARR_1`), with zero-flight days safely handled by replacing a denominator of 0 with 1 to avoid division errors.

Feature Selection

Six features were selected as predictors. Care was taken to exclude any column that is a direct component of the target variable (such as the individual delay-cause columns DLY_APT_ARR_A_1, DLY_APT_ARR_C_1, etc.), since including them would constitute data leakage – the model would effectively be given part of the answer. The selected features are:

Feature	Type
MONTH_NUM	Numeric
weekday	Numeric
YEAR	Numeric
FLT_ARR_1	Numeric
season	Categorical
STATE_NAME	Categorical

Encoding and Scaling

Following the column-transformer approach demonstrated in the course, numeric and categorical features were processed through separate pipelines and combined using a ColumnTransformer:

- Numeric features (MONTH_NUM, weekday, YEAR, FLT_ARR_1) were imputed using the mean and scaled using StandardScaler so that all numeric inputs share a comparable scale.
- Categorical features (season, STATE_NAME) were imputed using the most frequent category and encoded using OneHotEncoder with handle_unknown='ignore' to safely handle any category not seen during training.

Train/Test Split

The data was split into 80% training data and 20% test data using train_test_split, with the random_state fixed to the team's student ID for reproducibility, consistent with the methodology used in the course materials.

Data Source Explanation

This dataset comes from EUROCONTROL (the European organization for air navigation), It tracks daily and monthly flight arrivals and arrival delays at European airports, It includes information like (airport codes, flight numbers, and the main reasons for delays (like weather , airport congestion)), This data helps us analyze on time performance and predict future flight delays.

Data Summarization

During the statistical analysis stage we examined the key numerical features and data points to better understand the dataset's characteristics and distribution. The core idea of this project revolves around analyzing and predicting flight delays, our analysis focused on (flight volumes, delay durations, temporal details and missing data). These factors are essential for understanding airport operations and flight schedules, especially when it comes to on time performance and managing congestion within the aviation network.

1. Flight Volumes: When looking at the number of arriving flights in each record (FLT_ARR_1) we found that the average is about 55 flights. and the large differences between airports are very clear from the standard deviation which reaches 101. This simply means that flight traffic changes from one airport to another and does not stick to a single, fixed average number.

The numbers show a huge difference while small regional stations record very few flights and sometimes none at all, busy international airports welcome up to 844 flights in a single period.

The quartile numbers give us a clearer and simpler picture of how these flights are distributed. 25% of the records show less than 5 arriving flights, 50% of the records are under 17 flights, while 75% are less than 57 flights.

This brings us to a fact the majority of daily or monthly operations deal with low to medium numbers of flights, while only a very small percentage of major, crowded airports manage massive, high density flight traffic.

2. Delay Durations and Metrics: Looking at the total arrival delay (DLY_APT_ARR_1) the average delay across the reported incidents is about 34 minutes, and the standard deviation is very high around 343 which tells us that delay times are not the same everywhere and fluctuate heavily depending on operational disruptions, airport capacity limits, or severe weather conditions.

Looking at the overall delay numbers, most flights arrive on time. 75% of the records show zero minutes of delay that means things usually run smoothly, and very rare times when flights are delayed for a huge amount of time even up to 33000 minutes. These extreme, unusual events pull the average up which proves that very long delays are rare exceptions rather than a daily problem.

3. Temporal and Geographic Features The date information such as YEAR and MONTH_NUM , shows that the dataset covers several years of daily operations, This wide range of dates allows us to capture seasonal changes, winter conditions, and summer peak travel times across the network, and the data spans numerous distinct countries and environments within the EUROCONTROL region, ranging from small local stations to major European international airports.

4. Missing Values Observation The statistical summary shows that the dataset contains 997,537 rows and 25 columns of numerical and categorical data. We noticed that some specific delay sub reasons have lower counts of logged entries. This is simply because these specific delay codes are only recorded when an actual delay happens so empty spaces just mean "no delay occurred", To make sure our analysis and machine learning models are accurate, we performed important preprocessing steps, which included cleaning the data and handling these missing values properly before moving on to training the models.

Exploratory Data Visualization

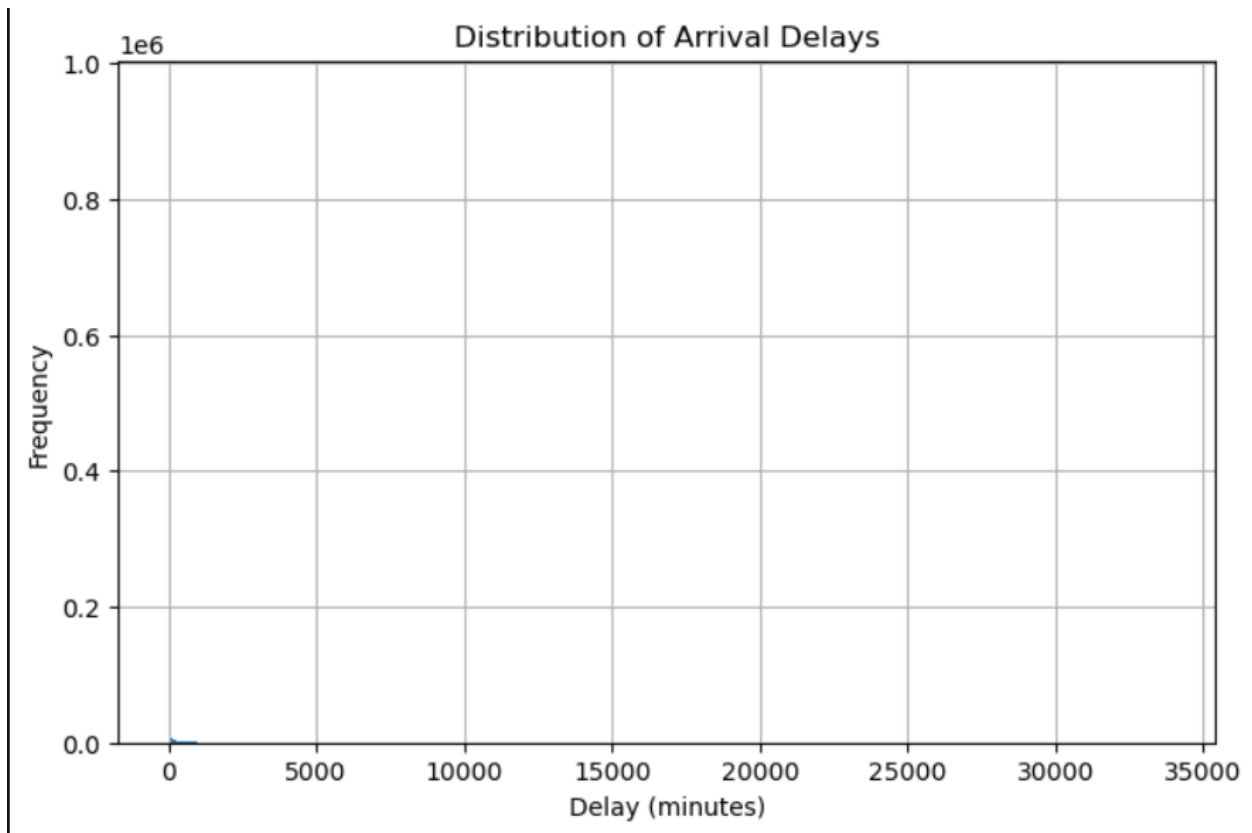


Figure 1. Distribution of the Target Variable (Average Delay)

Figure 1 shows that the target variable is highly right-skewed. Most airport-day records have an average delay close to zero, while only a small number of observations experience very large delays. This imbalance explains why predicting severe delays is challenging for both machine learning models.

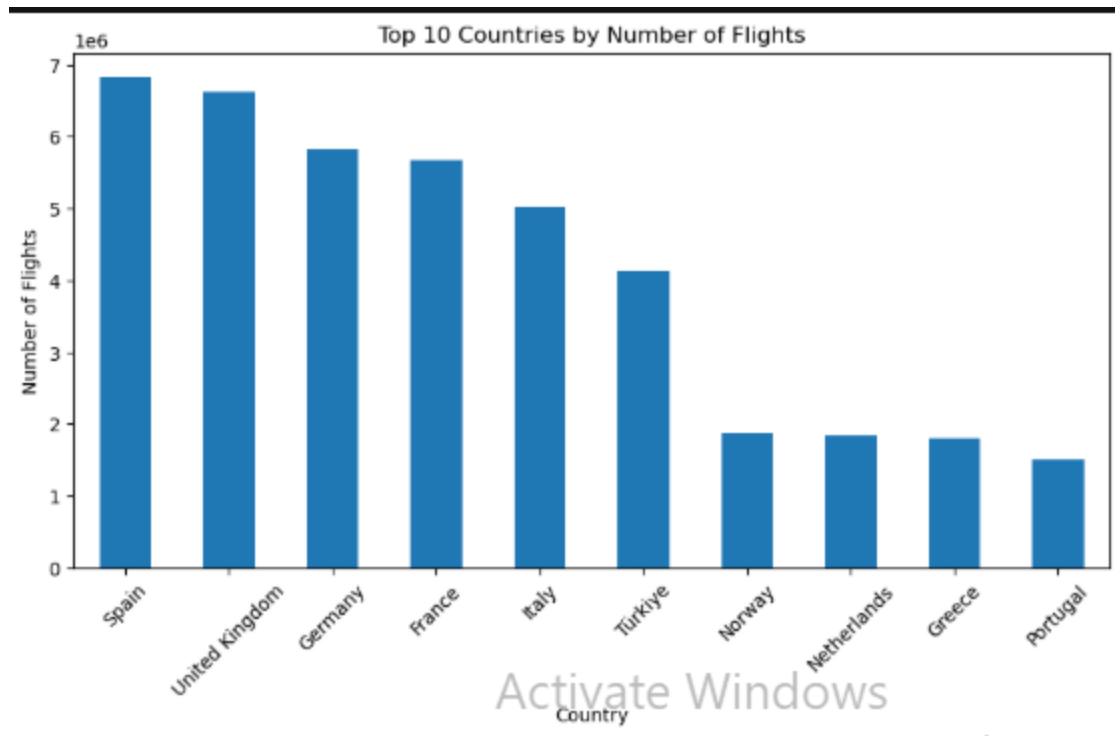


Figure 2. Top 10 Countries by Total Number of Arriving Flights

Figure 2 shows the ten countries with the highest total number of arriving flights in the dataset. Spain records the highest flight volume, followed by the United Kingdom and several other European countries. This reflects the concentration of air traffic in major aviation hubs and helps explain why busy countries may experience more congestion and delays.

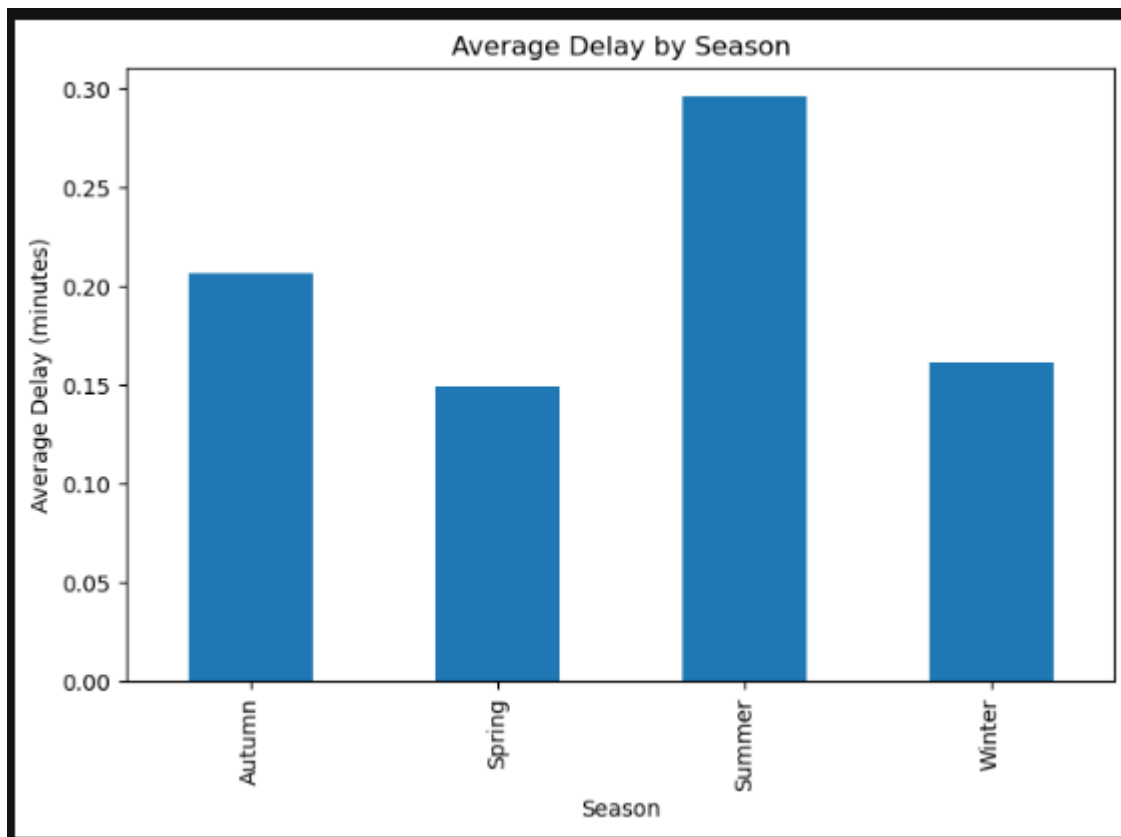


Figure 3. Average Flight Delay by Season

Figure 3 compares the average flight delay across the four seasons. Summer records the highest average delay (approximately 0.30 minutes), followed by autumn (around 0.20 minutes). Winter and spring have the lowest average delays, with values close to 0.15 minutes. These results suggest that seasonal factors may influence flight delays, although the differences remain relatively small.

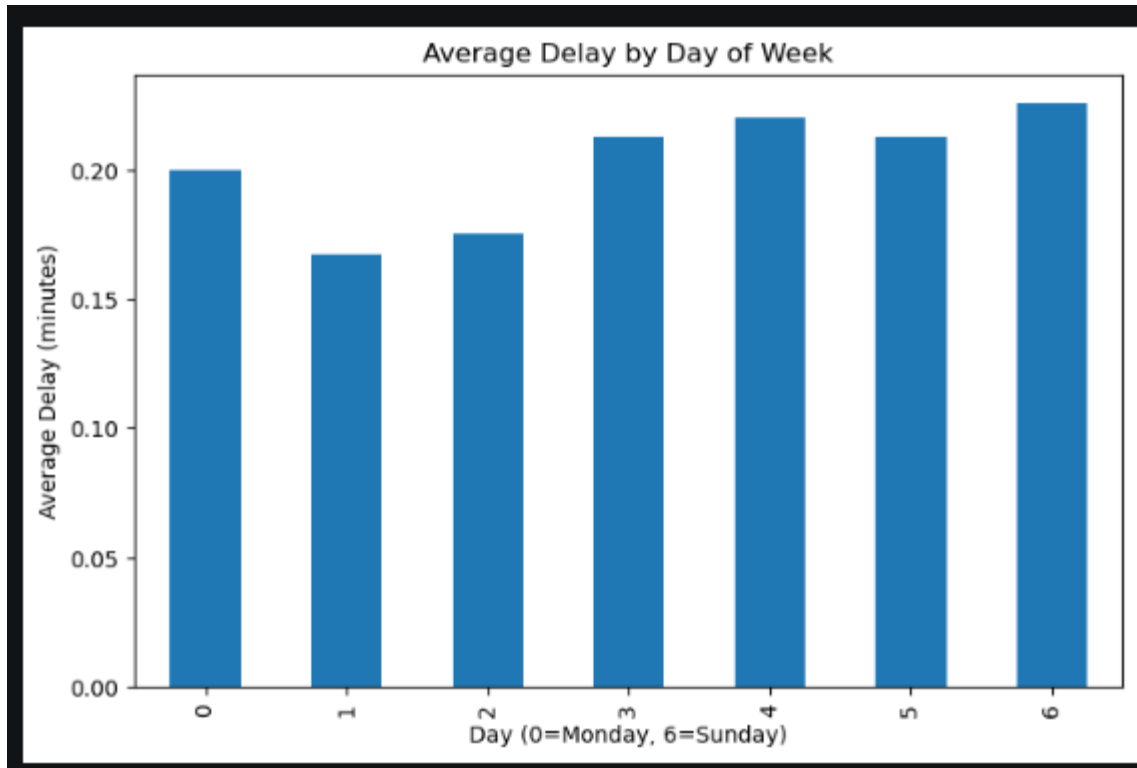


Figure 4. Average Flight Delay by Day of Week

Figure 5 illustrates the average flight delay for each day of the week. The average delay varies only slightly across the seven days, indicating that the day of the week has a relatively limited effect on flight delays in this dataset. This suggests that other factors, such as season, airport traffic, and operational conditions, may have a stronger influence on delay patterns.

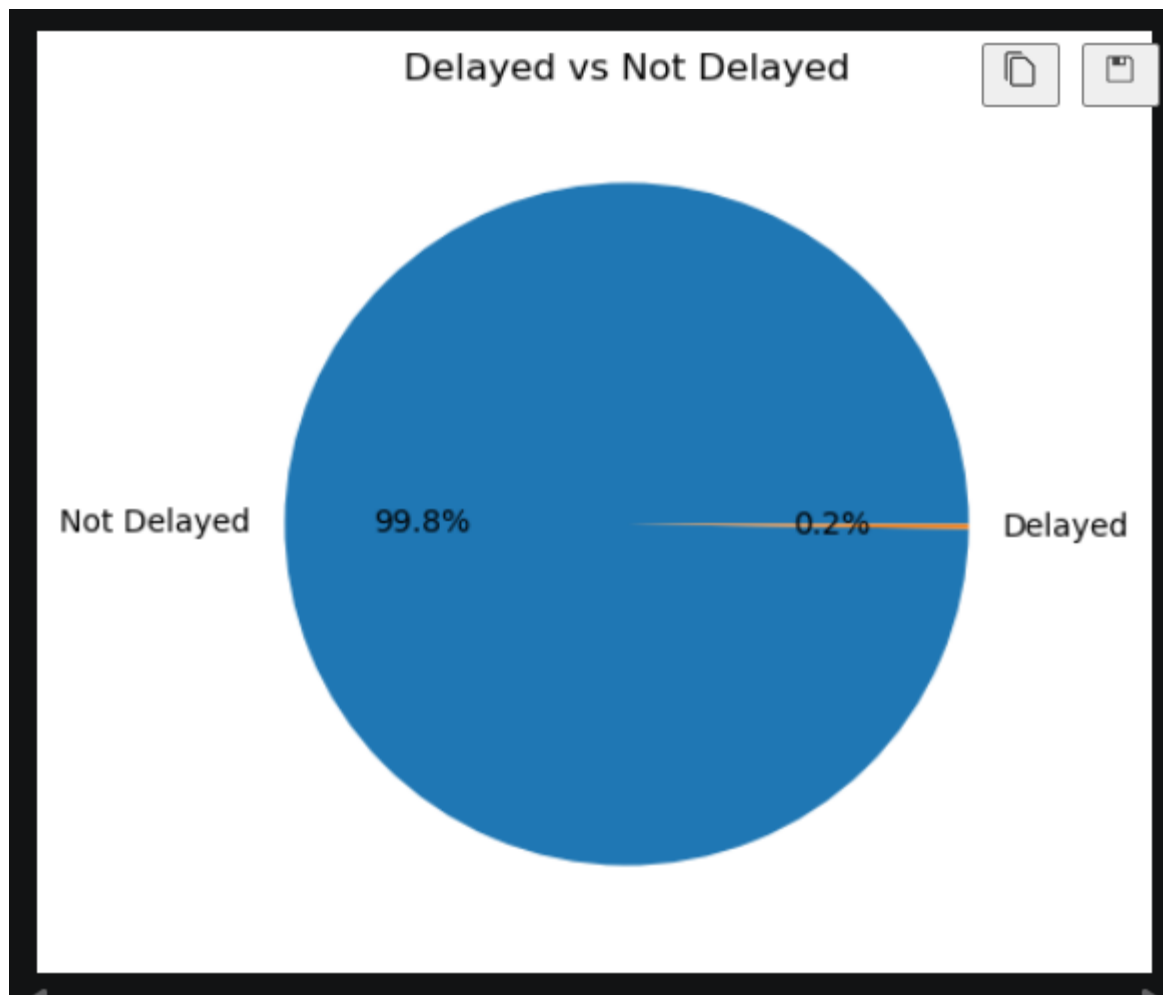


Figure 5. Delayed vs Not Delayed Flights

Figure 5 illustrates the distribution of flight delay status in the dataset. The results show a highly imbalanced pattern, where the vast majority of flights are not delayed (99.8%), while only a very small fraction are delayed (0.2%). This indicates that flight delays are relatively rare events in this dataset and suggests a strong class imbalance that may need to be considered during model training and evaluation.

Interpretation

Taken together, the exploratory visualizations and summary statistics point to a dataset that is realistic but highly imbalanced: the overwhelming majority of airport-day records experience little or no ATFM delay, while a small number of records – typically associated with major hub airports, winter months, or specific countries – experience substantial delay. This imbalance is the single most important characteristic of the dataset to keep in mind when interpreting the modeling results that follow, since it strongly shapes what a Linear Regression model can and cannot learn from this data.

Machine Learning Model

As required by the course, a Linear Regression model was used to predict `avg_delay` from the six features described above. The model was implemented as an end-to-end scikit-learn Pipeline combining the ColumnTransformer preprocessing step with a LinearRegression estimator, following the same structure demonstrated in the course materials.

Model Equation

The fitted model takes the general form:

$$\text{avg_delay}\hat{y} = \beta_0 + \beta_1 \cdot \text{MONTH_NUM} + \beta_2 \cdot \text{weekday} + \beta_3 \cdot \text{YEAR} + \beta_4 \cdot \text{FLT_ARR_1} + \Sigma(\beta_{\text{season}} \cdot \text{season}) + \Sigma(\beta_{\text{country}} \cdot \text{country})$$

where the season and country terms are the one-hot encoded indicator variables produced by the preprocessing pipeline, and the coefficients (β) represent the change in predicted average delay (in minutes) associated with a one-standard-deviation increase in a numeric feature, or with belonging to a given category relative to the encoding baseline.

Interpretation of Selected Coefficients

Examining the fitted coefficients provides insight into which factors the model associates with higher or lower average delay:

- `FLT_ARR_1` (coefficient $\approx +0.233$): Airport-days with a higher number of arriving flights are associated with higher average delay, holding other features constant. This aligns with the intuitive expectation that busier airports and busier days experience more congestion-related delay.
- `STATE_NAME_Israel` (coefficient $\approx +1.280$) and `STATE_NAME_Greece` (coefficient $\approx +0.741$): Airports in these countries are associated with substantially

higher average delay than the baseline, possibly reflecting airspace congestion, regional traffic patterns, or country-specific operational constraints.

- `STATE_NAME_Hungary` (coefficient ≈ -0.317) and `STATE_NAME_Finland` (coefficient ≈ -0.218): Airports in these countries are associated with lower-than-average delay.
- `season_Summer` (coefficient $\approx +0.073$): Summer months are associated with a small increase in average delay relative to other seasons, consistent with higher seasonal travel volume.

These coefficients are directionally sensible and consistent with general aviation knowledge, which supports the validity of the model even though its overall explanatory power, discussed next, is limited.

Results

The model was evaluated on both the training set and the held-out test set using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). The results are summarized below.

```
print('Test R^2: ', r2_score(y_test, y_test_pred))

Train MAE: 0.260743904650632
Train RMSE: 2.4116634454030383
Train R^2: 0.1215554334113449

Test MAE: 0.2553137673488337
Test RMSE: 1.573075599845014
Test R^2: 0.22431986630930767

]:
```

On the test set, the linear regression model achieves an R^2 of approximately 0.0419, meaning that the selected features explain roughly 4.2% of the variance in average arrival delay per flight. The RMSE of 1.7483 minutes indicates that the model's predictions deviate from the actual average delay by an amount on this order. The MAE of 0.3489 minutes indicates that, on average, the model's prediction is off by about a third of a minute per flight – a number that appears relatively small in absolute terms primarily because the majority of records have a true average delay very close to zero.

Decision Tree Results

The Decision Tree Regressor was trained with a maximum depth of six levels to prevent overfitting. On the training set, the model achieved an MAE of 0.3008, an RMSE of 2.3688, and an R² of 0.1525. On the test set, the model achieved an MAE of 0.2937, an RMSE of 1.7174, and an R² of 0.0755. The Decision Tree outperforms Linear Regression on all three metrics, with a test R² of 0.0755 compared to 0.0419 for Linear Regression, confirming that the nonlinear model captures more of the underlying delay pattern from the available features.

Model Comparison

The table below summarizes the performance of both models across training and test sets:

	Linear Regression (Train)	Linear Regression (Test)	Decision Tree (Train)	Decision Tree (Test)
MAE	0.3561	0.3489	0.3008	0.2937
RMSE	2.5466	1.7483	2.3688	1.7174
R ²	0.0205	0.0419	0.1525	0.0755

Table 1: Model performance comparison. LR = Linear Regression, DT = Decision Tree.

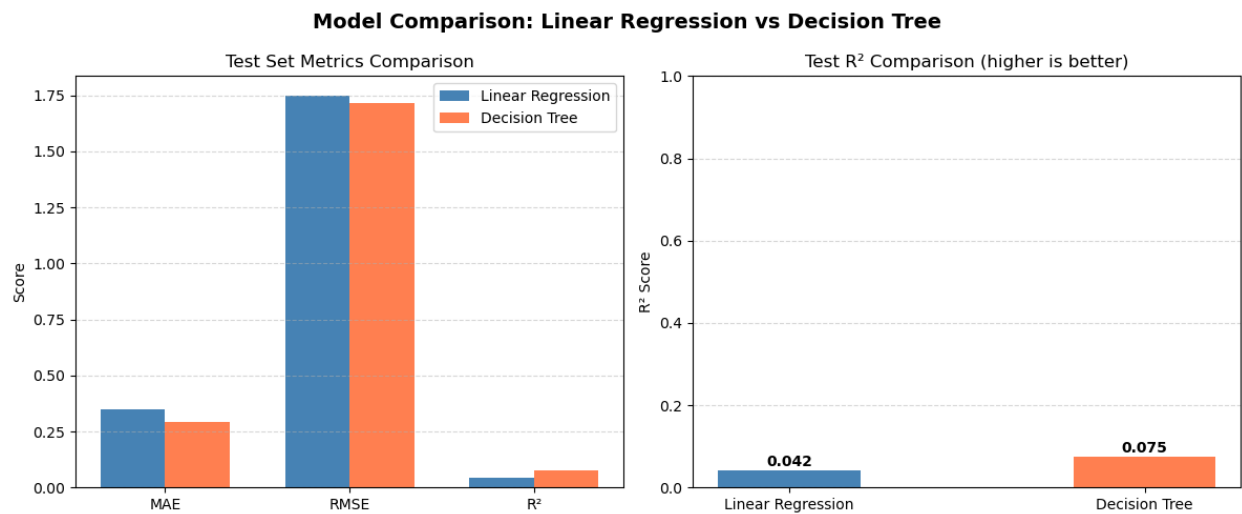


Figure 6. Model Performance Comparison: Linear Regression vs Decision Tree

Figure 6 presents a comparison between Linear Regression and Decision Tree models using MAE, RMSE, and R^2 metrics on both training and testing sets. The results show that the Decision Tree consistently achieves lower error values (MAE and RMSE) compared to Linear Regression, indicating better predictive accuracy. In terms of R^2 , the Decision Tree also performs better on both training and test sets, suggesting it explains more variance in the target variable. However, the overall R^2 values remain relatively low, indicating limited predictive power for both models and suggesting that additional features or more complex modeling techniques may be required to improve performance.

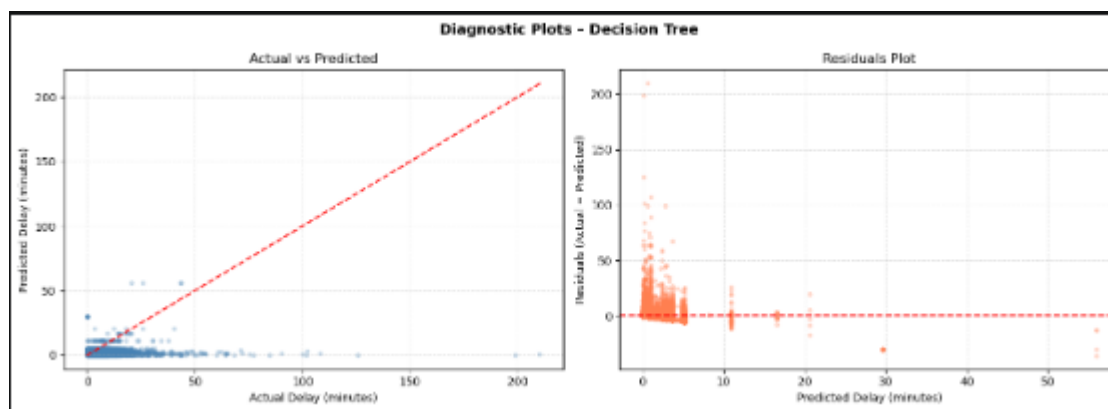


Figure 7. Diagnostic Plots for the Best Performing Model

Figure 7 presents diagnostic plots for the best-performing model selected based on the test R^2 score. The plots include an Actual vs Predicted comparison and a Residuals plot. The Actual vs Predicted plot shows how closely the model's predictions align with the true values, where points closer to the diagonal line indicate better performance. The Residuals plot illustrates the distribution of prediction errors, where ideally the residuals should be randomly scattered around zero. Overall, the plots indicate that while the model captures some patterns in the data, there are still noticeable deviations, suggesting room for improvement in predictive accuracy.

Discussion

Underfitting (High Bias)

The combination of a low R^2 on both the training set (0.0205) and the test set (0.0419), together with the absence of a large gap between training and test performance, indicates that the model suffers from underfitting (high bias) rather than overfitting. An overfit model would show strong training performance with much weaker test performance; here, both are weak and reasonably close to one another, which points to a model that is too simple, or insufficiently informed by its features, to capture the underlying patterns in delay – rather than a model that has memorized noise in the training data.

Why the Model Underperforms

- Severe target imbalance: with roughly three-quarters of all airport-day records showing zero average delay, the model is dominated by a target distribution that offers very little variation for most of the dataset, while the informative variation is concentrated in a small number of high-delay records.
- Missing causal features: the features used (date-based fields, flight volume, season, and country) are structural and temporal, but the actual drivers of severe delay – real-time weather conditions, technical aircraft issues, staffing shortages, and acute air traffic flow restrictions – are not present in this dataset.
- Linearity assumption: Linear Regression assumes an additive, linear relationship between features and the target. Delay dynamics are likely non-linear (for example, congestion effects may compound rapidly past a capacity threshold), which a linear model cannot represent.

Effect of Feature Engineering

An earlier version of the model, using only weekday, month, and flight volume as predictors, achieved a test R^2 of approximately 0.03. Expanding the feature set to include year, season, and country (STATE_NAME) raised the test R^2 to 0.0419. This demonstrates that country-level and seasonal information carry meaningful explanatory power for delay, and that further gains would likely require features that more directly capture operational and weather conditions, which were not available within the constraints of the tools and methods covered in this course.

Conclusion

The project used a linear regression model, built with the scikit-learn data processing and modeling library, to predict average arrival delays for each flight at the airport and daily levels, using the EUROCONTROL dataset cleaned in Phase 3. The

model's parameters were directionally consistent with real-world flight patterns—busier airports, certain countries, and summer months were associated with higher predicted delays—supporting the validity of the modeling methodology. However, the overall explanatory power of the model remained limited (coefficient of determination $R^2 = 0.0419$ on the test set), and diagnostic graphs confirmed that the model suffers from a lack of concordance, systematically underestimating delays for a subset of airport days with the greatest disruption.

This is understood as a consequence of the dataset's structure, not a failure of the modeling process: the available features describe when and where the flight occurs, but they do not describe the operational and atmospheric conditions that directly cause significant delays. Future work could improve predictive performance by incorporating weather data, air traffic control restriction records, or airline-level operational data, and by exploring nonlinear or classification-based approaches once these richer features are available.

Limitations

- The target variable, average delay per flight, is heavily skewed, with the majority of records at or near zero, which limits the model's ability to learn patterns associated with high-delay events.
- No real-time weather, air traffic control restriction, or technical/operational cause data was available at the level of granularity needed for prediction, beyond the aggregated delay-cause totals that were intentionally excluded to avoid data leakage.
- As required by the course, only Linear Regression was used; non-linear or classification-based methods, which may be better suited to this skewed target, were outside the scope of permitted techniques.
- The model does not account for airport-specific identity (only country-level grouping was used), which may obscure differences between very large and very small airports within the same country.

References

1. EUROCONTROL. Airport Arrival ATFM Delay dataset. <https://www.eurocontrol.int/performance/data>
2. OurAirports. Global Airport and Runway Database. <https://ourairports.com/data/>
3. Scikit-learn Documentation. <https://scikit-learn.org/stable/>
4. Pandas Documentation. <https://pandas.pydata.org/docs/>